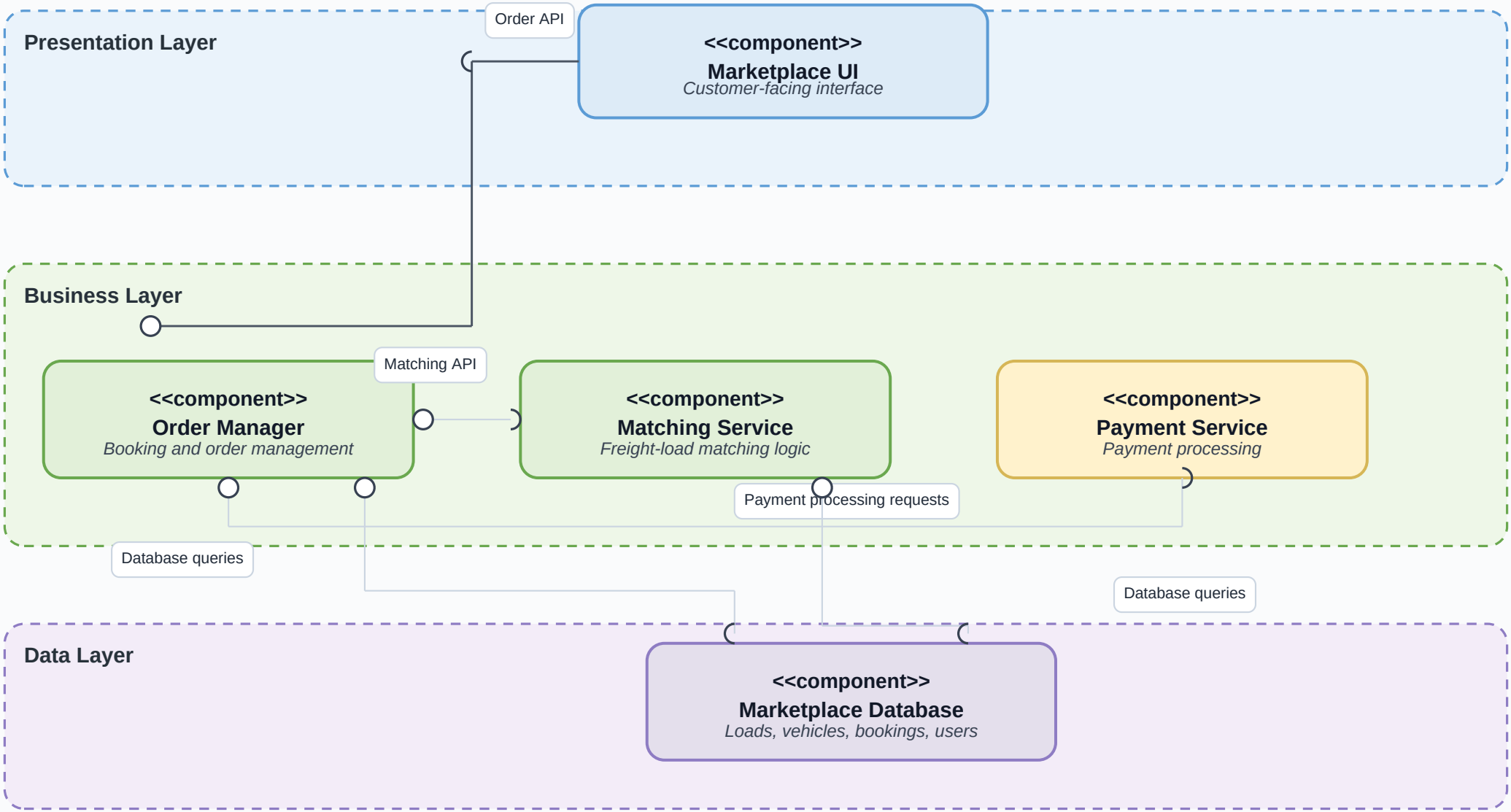
LAB 3 - COMPONENT DIAGRAM

Inter-City Freight Load Matching Marketplace

Atharv Kulkarni | SRN: PES2UG24CS092 | Selected Architecture: Layered



UML notation: provided interface = ball (circle) | required interface = socket (semicircle) | labelled connectors show service/data flow

LAB 3 - ARCHITECTURAL JUSTIFICATION

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Architecture Selection: We chose **Layered Architecture** for the Inter-City Freight Load Matching Marketplace.

The handout asks students to compare Layered, Microservices and Client-Server styles and select the most suitable style for the assigned scenario. For this marketplace lab, Layered Architecture gives a clear separation between the customer-facing interface, business logic and data storage, while keeping the component interactions straightforward.

Architectural Choice - Two Scenario-Related Reasons

1. Clear separation of responsibilities. The Marketplace UI handles user interaction; the Order Manager, Matching Service and Payment Service handle business functions; and the Marketplace Database stores persistent marketplace information. This separation makes the component responsibilities easier to understand and maintain.

2. Simpler integration for the core workflow. Freight posting, matching, booking/order handling and payment processing require predictable interactions between a small set of components. A layered structure provides direct, understandable dependencies without the operational complexity associated with independently deployed microservices.

Security Advantage

Security responsibilities can be concentrated in the business layer: the Order Manager can enforce authorization and validate booking requests before allowing payment or data operations. Keeping database access behind business components also reduces direct access from the customer-facing layer.

Performance Benefit

The layered design avoids unnecessary distribution overhead for a lab-sized deployment. Separating the matching logic into its own Matching Service also allows that computation to be optimized without changing the user interface.

Component Identification

Layer	Component	Purpose
Presentation	Marketplace UI	Customer-facing interaction for shippers and transporters.
Business	Order Manager	Coordinates freight/order and booking operations.
Business	Matching Service	Evaluates and returns suitable freight/transport matches.
Business	Payment Service	Handles payment processing requests.
Data	Marketplace Database	Stores users, freight, vehicle/trip and booking data.

Interfaces shown: Order API, Matching API, Payment API and Database Queries (including a Database Queries connection from the Matching Service).